

Food Waste Reduction and Donation Management System

Modification Based on Literature Survey

SE LAB 03 A

Submitted By:

Rudrakshi Chincholkar (241071014)

Nihira Neralwar (241071049)

April 5, 2026

Section

1	Problem Statement	2
2	Refined Core Problem	2
3	Expanded Functional Requirements	2
3.1	Donation & Food Management	2
3.2	Pickup & Delivery Operations	2
3.3	Food Safety & Inspection	3
3.4	Administration & System Management	3
3.5	Feedback & Transparency	3
4	System Actors	3
5	Use Cases	3
6	Use Case Diagram	4
7	Conclusion	4

1 Problem Statement

In modern society, a significant quantity of edible food is wasted daily by restaurants, hotels, supermarkets, and households, while many individuals continue to face hunger and food insecurity. Existing food donation practices are largely manual, uncoordinated, and lack structured monitoring mechanisms.

Current challenges include:

- **Lack of Centralized Coordination:** Donors, NGOs, and volunteers operate without a unified platform.
- **Inefficient Pickup Management:** Absence of structured request handling causes delays and food spoilage.
- **Food Safety Concerns:** No standardized inspection or safety monitoring process.
- **Limited Transparency:** Stakeholders cannot track donations throughout the delivery lifecycle.
- **Poor System Oversight:** Administrators lack analytics and monitoring tools.

2 Refined Core Problem

The core problem is the absence of a structured digital ecosystem that enables:

- Safe food donation management
- Efficient pickup and delivery coordination
- Volunteer and delivery partner management
- Food inspection and safety compliance
- Real-time tracking and transparency
- Administrative monitoring and reporting

3 Expanded Functional Requirements

3.1 Donation & Food Management

- Users can Register/Login into the system.
- Food donors can Add/Edit/Delete Food Requests.
- Donors must follow Food Safety Guidelines before listing food.
- NGOs can View Available Donations and accept suitable food.

3.2 Pickup & Delivery Operations

Volunteers can:

- Set Availability and View Pickup Requests.

- Accept Pickup Tasks and Update Pickup Status.

Delivery Partners can:

- Accept Delivery Tasks and Pick Up Food.
- Deliver Food to NGOs and Update Delivery Status.

3.3 Food Safety & Inspection

- Define Food Safety Rules.
- Inspect Food Donations and Flag Unsafe Food Items.

3.4 Administration & System Management

- Manage Users, Volunteers, and Delivery Partners.
- Assign Food Inspectors and Penalize violations.
- Monitor System Activity and Generate Reports.

3.5 Feedback & Transparency

- Users can provide Feedback and Ratings.
- Donation lifecycle tracking ensures accountability.

4 System Actors

1. **Food Donor:** Adds donation requests and follows safety guidelines.
2. **NGO:** Views and accepts donations; handles unsafe cases.
3. **Volunteer:** Manages local collection and status updates.
4. **Delivery Partner:** Professional logistics and transportation.
5. **Food Safety Authority:** Sets standards and inspects food items.
6. **System Administrator:** Monitors activity and manages the database.

5 Use Cases

The system supports the following primary use cases:

- Register / Login
- Add/Edit/Delete Food Request
- Follow Food Safety Guidelines
- Accept Pickup/Delivery Tasks
- Inspect Food Donations
- Track Donation Lifecycle
- Generate Reports

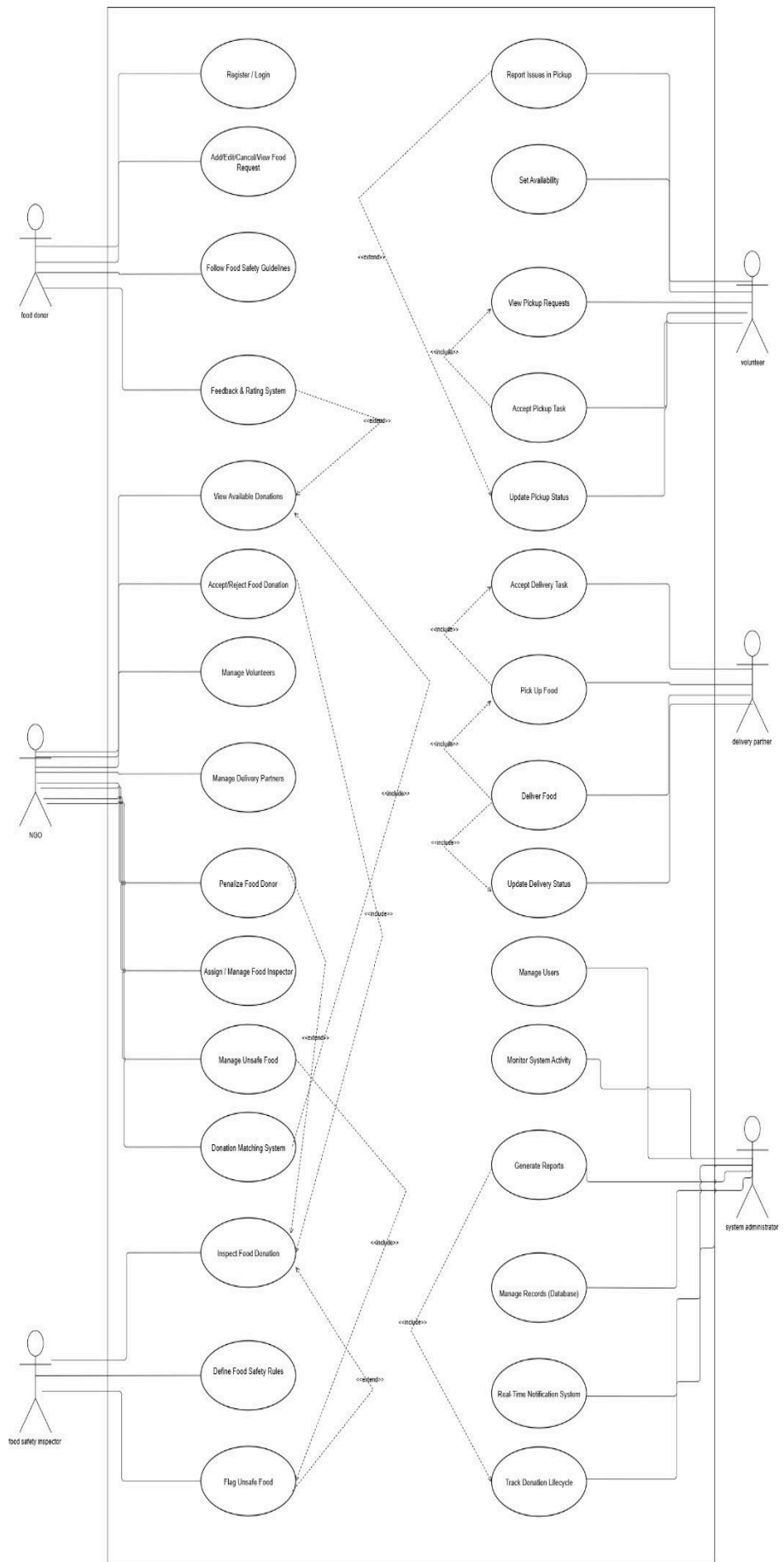


Figure 1: UseCase diagram

6 Use Case Diagram

The use case diagram illustrates interactions between all actors. It shows how donations move from creation to inspection, pickup, delivery, and final monitoring within the system.

7 Conclusion

The proposed Food Waste Reduction and Donation Management System provides a structured and transparent digital platform for managing surplus food redistribution. By integrating donation management, volunteer coordination, and food safety inspection, the system ensures efficient logistics and safe food handling.